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RaceDay — API Endpoint Plan

Roles referenced below: None (public, no login required), Any (any logged-in user, either role), Participant, Organizer.

Authentication

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/auth/register	Registers a new account as either a Participant or an Organiser.	None	{name, email, password, role, phone}	201 Created – account created, returns user ID · 409 Conflict – email already registered
POST	/api/auth/login	Authenticates a user and returns an access token for subsequent requests.	None	{email, password}	200 OK – returns auth token and role · 401 Unauthorized – invalid credentials

User Profile

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/users/me	Returns the profile of the currently logged-in user (Participant or Organiser).	Any	None	200 OK – returns profile data · 401 Unauthorized – no valid token
PUT	/api/users/me	Updates the logged-in user's own profile details.	Any	{name, phone, emergencyContact } (Participant) or {name, phone, organisationName } (Organiser)	200 OK – profile updated · 400 Bad Request – invalid fields

Events

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events	Lists all upcoming events, so Participants can browse without logging in.	None	None	200 OK – returns array of events
GET	/api/events/{id}	Returns full details for a single event, including its venue and categories.	None	None	200 OK – returns event details · 404 Not Found – event does not exist
POST	/api/events	Creates a new event under the logged-in Organizer's account.	Organiser	{ eventName, eventType, eventDate, venueId, description }	201 Created – event created · 400 Bad Request – missing/invalid fields

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
PUT	/api/events/{id}	Updates the event's details. Restricted to the Organiser who created it.	Organiser (owner)	{ eventName, eventType, eventDate, venueId, description }	200 OK – event updated · 403 Forbidden – not the event owner · 404 Not Found
DELETE	/api/events/{id}	Cancels/removes an event. Restricted if Categories with existing Entries exist.	Organiser (owner)	None	200 OK – event deleted · 409 Conflict – event has existing entries

Categories

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events/{eventId}/categories	Lists all categories available under a specific event, with route and pricing info.	None	None	200 OK – returns array of categories · 404 Not Found – event does not exist
POST	/api/events/{eventId}/categories	Adds a new category (e.g. 10km, 21km) to an event, linked to a Route.	Organiser (owner)	{ categoryName, routeId, minAge, entryFee }	201 Created – category created · 404 Not Found – event or route does not exist
PUT	/api/categories/{id}	Updates a category's details (name, route, age limit, fee).	Organiser (owner)	{ categoryName, routeId, minAge, entryFee }	200 OK – category updated · 403 Forbidden – not the event owner
DELETE	/api/categories/{id}	Removes a category. Restricted if Entries already exist against it.	Organiser (owner)	None	200 OK – category deleted · 409 Conflict – category has existing entries

Event Enrolments (Entries)

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/categories/{categoryId}/entries	Enter the logged-in Participant into a specific category. Validates minimum age and duplicate entry rules.	Participant	{ paymentStatus }	201 Created – entry created, returns race number · 400 Bad Request – below minimum age · 409 Conflict – already entered this category

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/users/me/entries	Returns all of the logged-in Participant's entries, past and upcoming.	Participant	None	200 OK – returns array of entries
GET	/api/entries/{id}	Returns details of a single entry.	Participant (owner) or Organizer (event owner)	None	200 OK – returns entry details · 403 Forbidden – not owner · 404 Not Found
DELETE	/api/entries/{id}	Cancels an entry before the event takes place.	Participant (owner)	None	200 OK – entry cancelled · 409 Conflict – event already took place

Results

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/entries/{id}/result	Captures a result for a completed entry. Only allowed once the event date has passed.	Organizer (event owner)	{ finishTime, position, status }	201 Created – result recorded · 400 Bad Request – event has not occurred yet · 409 Conflict – result already exists for this entry
GET	/api/entries/{id}/result	Retrieves the result for a specific entry, if one exists.	Any	None	200 OK – returns result · 404 Not Found – no result recorded yet
GET	/api/users/me/results	Returns the logged-in Participant's full performance history across all past events.	Participant	None	200 OK – returns array of results
PUT	/api/results/{id}	Corrects a previously captured result.	Organizer (event owner)	{ finishTime, position, status }	200 OK – result updated · 403 Forbidden – not the event owner

Additional Endpoints (Weather & Route Info)

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events/{id}/weather	Returns live weather for the event's venue, using the Venue's stored coordinates, so Participants can prepare for race day.	None	None	200 OK – returns live weather data · 503 Service Unavailable – external weather API unreachable

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/routes/{id}	Returns route details (distance, elevation, map link) for a given category's course.	None	None	200 OK – returns route details · 404 Not Found